Milestone1

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Using simulated data modelled on OECD unemployment statistics, this project plans to compare tertiary-educated foreign-born and native-born women across more and less developed regions.

Unemployment Rate among Highly Educated Immigrant Women by the Country of Destination

Introduction

Questions

Immigrants with a university degree are often expected to secure stable positions in the labor market, yet a degree does not guarantee the same outcome across nations. Immigrants may face disadvantages depend on where they settle and on when they arrived. For this research, I focus on highly educated female immigrants and ask: **Does the unemployment rate of tertiary-educated foreign-born women vary by the development level of the destination country? Specifically, is it lower if immigrant women migrate to more developed countries?**

As more developed countries may have more established systems for recognizing foreign qualifications, I expect highly educated foreign-born women to experience lower unemployment in Northern America, Oceania, Western Europe and Northern Europe, which I group as "more developed" regions, than in Latin America, Southern Europe and Eastern Europe, which I group as "less developed" regions.

To make the analysis more robust, I also account for general labor market conditions. A low unemployment rate among immigrants may simply reflect a strong labor market that benefits everyone. To separate the two, I also compare each foreign-born group with native-born women of the same region and education level, and examine the gap between them (foreign-born minus native-born unemployment rate). If immigrant women are better off in more developed regions for reasons beyond general labor market conditions, I expect this gap to be smaller in more developed regions than in less developed ones.

Data

The analysis was conducted in R (R Core Team 2021) (Alexander 2023). For this project, I plan to using **2023 OECD data on unemployment rates** (OECD 2023). The data come from the OECD Data Explorer (DSD_EAG_LSO_EA@DF_LSO_NEAC_UNEMP_MIGR). Each row is a subgroup in one country, defined by sex, age group, education level, place of birth (foreign-born or native-born) and, for the foreign-born, age at arrival (before age 16, or 16 and older). The outcome variable is the unemployment rate, measured as a percentage of the subgroup's labor force. I plan to keep women with tertiary education, group countries into regions, and then into "more developed" and "less developed" regions. I compare foreign-born women **by region**, and compare them with native-born women of the same region and education level. I exclude Israel because it is difficult to classify as either a more or less developed region, and I exclude the OECD average because it is not a country.

Limitations

Although the OECD provides detailed data, it does not report immigrants' country of origin. Where immigrants come from may affect whether their educational credentials are recognized in the country of destination, and therefore their unemployment rate. As a result, differences across destination countries in this analysis may partly reflect differences in the origins of their immigrant populations, rather than characteristics of the destination countries themselves.

Expected Results

I will treat the hypotheses as supported if, for both arrival-age groups, the mean unemployment rate is lower in more developed than in less developed regions by about 1 percentage point or more. Moreover, the gap between foreign-born and native-born women (foreign-born minus native-born) is smaller in more developed regions than in less developed regions by about 1 percentage point or more. If the differences are small, or are inconsistent across groups, I will conclude that there is no clear pattern.

```
library(tidyverse)
library(tinytex)
library(knitr)
```

```
raw_set <- read_csv(
  "OECD_raw_v2.csv",
  show_col_types = FALSE
)
```

PART I: Simulating Data

```
set.seed(916)

#Simulate the structure of the real data, based on the data of OECD
country_region <- tibble(
  country = c("Canada", "United States",
              "Mexico", "Costa Rica", "Colombia", "Chile",
              "Australia", "New Zealand",
              "Austria", "Belgium", "France", "Germany", "Luxembourg",
              "Netherlands", "Switzerland",
              "Estonia", "Ireland", "Latvia", "Lithuania", "Norway",
              "Sweden", "United Kingdom",
              "Greece", "Italy", "Portugal", "Slovenia", "Spain",
              "Bulgaria", "Czechia", "Hungary", "Poland", "Romania",
              "Slovak Republic"),
  region  = c(rep("Northern America", 2),
              rep("Latin America", 4),
              rep("Oceania", 2),
              rep("Western Europe", 7),
              rep("Northern Europe", 7),
              rep("Southern Europe", 5),
              rep("Eastern Europe", 6))
)

head(country_region)
```

```
# A tibble: 6 x 2
  country       region
  <chr>         <chr>
1 Canada        Northern America
2 United States Northern America
3 Mexico        Latin America
4 Costa Rica    Latin America
5 Colombia      Latin America
6 Chile         Latin America
```

```
#add characteristics
n_countries <- length(country_region$country)

simulated_ppl <- tibble(
```

```
  country = rep(country_region$country, each = 18),
  sex = rep(rep(c("Female", "Male"), each = 9), times = n_countries),
  education = rep(rep(c("Below upper secondary",
                        "Upper secondary / post-secondary non-tertiary",
                        "Tertiary"), each = 3),
                  times = 2 * n_countries),
  birth_place = rep(c("Foreign-born", "Foreign-born", "Native-born"),
                    times = 6 * n_countries),
)

head(simulated_ppl)
```

```
# A tibble: 6 x 4
  country sex    education                                    birth_place
  <chr>   <chr>  <chr>                                        <chr>
1 Canada  Female Below upper secondary                        Foreign-born
2 Canada  Female Below upper secondary                        Foreign-born
3 Canada  Female Below upper secondary                        Native-born
4 Canada  Female Upper secondary / post-secondary non-tertiary Foreign-born
5 Canada  Female Upper secondary / post-secondary non-tertiary Foreign-born
6 Canada  Female Upper secondary / post-secondary non-tertiary Native-born
```

```
#add region and development group
simulated_groups <-
  simulated_ppl |>
  left_join(country_region, by = "country") |>
  mutate(
    development_group = case_match(
      region,
      c("Northern America", "Oceania", "Western Europe", "Northern Europe") ~ "More developed",
      c("Latin America", "Southern Europe", "Eastern Europe") ~ "Less developed",
      .default = NA_character_
    )
  )

head(simulated_groups)
```

```
# A tibble: 6 x 6
  country sex    education                 birth_place region development_group
  <chr>   <chr>  <chr>                     <chr>       <chr>  <chr>
1 Canada  Female Below upper secondary     Foreign-bo~ North~ More developed
```

```
2 Canada  Female Below upper secondary      Foreign-bo~ North~ More developed
3 Canada  Female Below upper secondary      Native-born North~ More developed
4 Canada  Female Upper secondary / post-se~ Foreign-bo~ North~ More developed
5 Canada  Female Upper secondary / post-se~ Foreign-bo~ North~ More developed
6 Canada  Female Upper secondary / post-se~ Native-born North~ More developed
```

```
#add unemployment rate
simulated_rates <-
  simulated_groups |>
  mutate(
    unemployment_rate = sample(seq(1, 15, by = 0.1),
                               size = n(),
                               replace = TRUE)
  )

head(simulated_rates)
```

```
# A tibble: 6 x 7
  country sex   education birth_place region development_group unemployment_rate
  <chr>   <chr> <chr>     <chr>       <chr>  <chr>                          <dbl>
1 Canada  Fema~ Below up~ Foreign-bo~ North~ More developed                  8.8
2 Canada  Fema~ Below up~ Foreign-bo~ North~ More developed                  9.9
3 Canada  Fema~ Below up~ Native-born North~ More developed                 11
4 Canada  Fema~ Upper se~ Foreign-bo~ North~ More developed                  5.6
5 Canada  Fema~ Upper se~ Foreign-bo~ North~ More developed                  5.2
6 Canada  Fema~ Upper se~ Native-born North~ More developed                  6.7
```

```
simulated_rates |>
  summarise(min_rate =
            min(unemployment_rate),
            max_rate = max(unemployment_rate))
```

```
# A tibble: 1 x 2
  min_rate max_rate
     <dbl>    <dbl>
1        1     14.9
```

```
simulated_data <-
  simulated_rates |>
  select(region, development_group, sex, education, birth_place, unemployment_rate)
```

```r
women_tertiary_simulated <-
  simulated_data |>
  filter(sex == "Female",
         education == "Tertiary") |>
  drop_na(development_group, unemployment_rate)

#only foreign-born women for the mean
mean_by_group <-
  women_tertiary_simulated |>
  filter(birth_place == "Foreign-born") |>
  group_by(development_group) |>
  summarise(mean_rate = mean(unemployment_rate))

mean_by_group
```

```
# A tibble: 2 x 2
  development_group mean_rate
  <chr>                <dbl>
1 Less developed        7.33
2 More developed        8.33
```

```r
#mean rate of native-born women
native_born_mean <-
  women_tertiary_simulated |>
  filter(birth_place == "Native-born") |>
  group_by(region, development_group) |>
  summarise(native_born_rate = mean(unemployment_rate), .groups = "drop")

native_born_mean
```

```
# A tibble: 7 x 3
  region           development_group native_born_rate
  <chr>            <chr>                        <dbl>
1 Eastern Europe   Less developed                6.05
2 Latin America    Less developed                6.75
3 Northern America More developed               11.6
4 Northern Europe  More developed                7.99
5 Oceania          More developed                4.9
6 Southern Europe  Less developed                7.64
7 Western Europe   More developed                8.43
```

```r
#mean rate of foreign-born women
foreign_born_mean <-
  women_tertiary_simulated |>
  filter(birth_place == "Foreign-born") |>
  group_by(region, development_group) |>
  summarise(foreign_born_rate = mean(unemployment_rate), .groups = "drop")

foreign_born_mean
```

```
# A tibble: 7 x 3
  region           development_group foreign_born_rate
  <chr>            <chr>                         <dbl>
1 Eastern Europe   Less developed                 8.12
2 Latin America    Less developed                 6.65
3 Northern America More developed                 7.55
4 Northern Europe  More developed                 8.96
5 Oceania          More developed                 7.7
6 Southern Europe  Less developed                 6.91
7 Western Europe   More developed                 8.09
```

```r
#calculate the gap by regions
gap_by_region <-
  women_tertiary_simulated |>
  group_by(region, development_group) |>
  summarise(
    native_born_rate = mean(unemployment_rate[birth_place == "Native-born"]),
    foreign_born_rate = mean(unemployment_rate[birth_place == "Foreign-born"]),
    .groups = "drop"
  ) |>
  mutate(gap = foreign_born_rate - native_born_rate)

gap_by_region
```

```
# A tibble: 7 x 5
  region           development_group native_born_rate foreign_born_rate     gap
  <chr>            <chr>                        <dbl>              <dbl>   <dbl>
1 Eastern Europe   Less developed                6.05               8.12  2.07
2 Latin America    Less developed                6.75               6.65 -
0.1000
3 Northern America More developed               11.6                7.55 -4
4 Northern Europe  More developed                7.99               8.96  0.979
```

```
5 Oceania          More developed               4.9              7.7   2.8
6 Southern Europe  Less developed               7.64             6.91 -
0.730
7 Western Europe   More developed               8.43             8.09 -
0.336
```

```
#calculate the gap by development groups
#The group means average the gaps of regions equally, without weighting by the number of cou
gap_by_development <- gap_by_region |>
  group_by(development_group) |>
  summarise(mean_gap = mean(gap))

gap_by_development
```

```
# A tibble: 2 x 2
  development_group mean_gap
  <chr>                <dbl>
1 Less developed       0.415
2 More developed      -0.139
```

Table 1

Table 1 shows 10 randomly selected rows of the simulated data, where each row is a subgroup in one country, defined by region and place of birth, together with its simulated unemployment rate.

```
simulated_data |>
  head(10) |>
  select(region, development_group, birth_place, unemployment_rate) |>
  kable(
    col.names = c("Region", "Group", "Place of birth", "Unemployment rate (%)"),
    caption = "Simulated data"
  )
```

Table 1: Simulated data

Region	Group	Place of birth	Unemployment rate (%)
Northern America	More developed	Foreign-born	8.8
Northern America	More developed	Foreign-born	9.9
Northern America	More developed	Native-born	11.0

Region	Group	Place of birth	Unemployment rate (%)
Northern America	More developed	Foreign-born	5.6
Northern America	More developed	Foreign-born	5.2
Northern America	More developed	Native-born	6.7
Northern America	More developed	Foreign-born	9.5
Northern America	More developed	Foreign-born	3.0
Northern America	More developed	Native-born	9.2
Northern America	More developed	Foreign-born	3.1

PART II: Graph

The following figures use the simulated data to show what the patterns would look like if the data followed each hypothesis.

Figure 1

Figure 1 compare the simulated mean unemployment rate of tertiary-educated foreign-born women in more and less developed regions.

```
#H1a
mean_by_group |>
  ggplot(aes(x = development_group, y = mean_rate)) +
  geom_col() +
  labs(x = NULL,
       y = "Mean unemployment rate",
       title = "Figure 1: Unemployment Rate of Tertiary-Educated\nForeign-Born Women by devel
  theme(plot.title = element_text(size = 12))
```

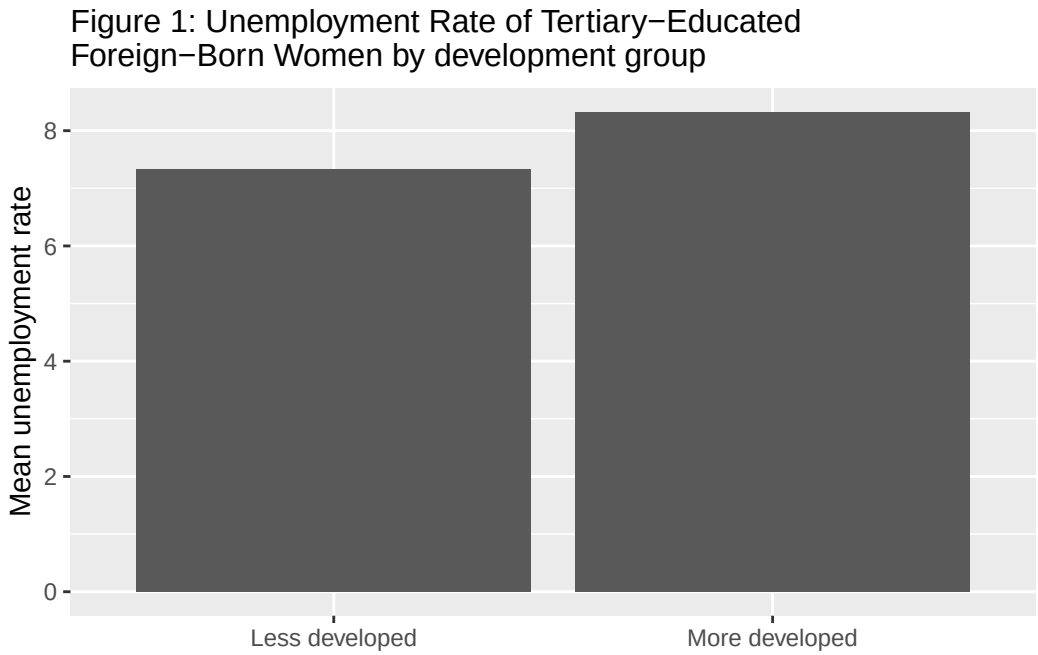


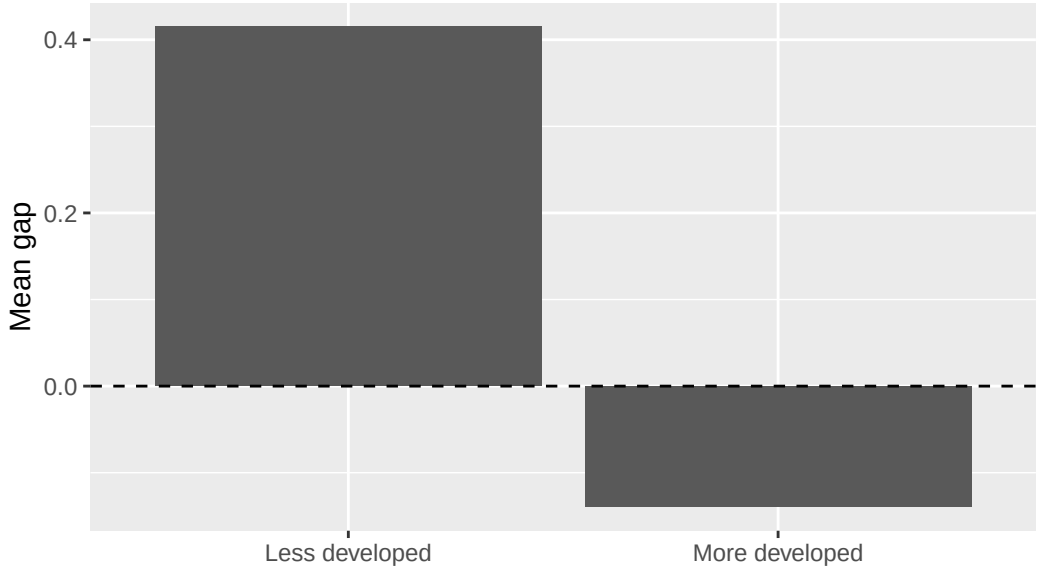
Figure 2

Figure 2 compares the simulated mean gap in unemployment rate between foreign-born women in more and less developed regions.

(The group means average the gaps of regions equally, without weighting by the number of countries or by population size.)

```
gap_by_development |>
  ggplot(aes(x = development_group, y = mean_gap)) +
  geom_col() +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(x = NULL,
       y = "Mean gap",
       title = "Figure 2: Unemployment rate gap between foreign-born and native-born\ntertia
  theme(plot.title = element_text(size = 12))
```

Figure 2: Unemployment rate gap between foreign−born and native-tertiary−educated women, by development group



Reference

Alexander, Rohan. 2023. *Telling Stories with Data*. Chapman; Hall/CRC. https://tellingstorieswithdata.com.

OECD. 2023. *DSD_EAG_LSO_EA@DF_LSO_NEAC_UNEMP_MIGR*. OECD Data Explorer; OECD. https://data-explorer.oecd.org/vis?lc=en&df%5Bds%5D=dsDisseminateFinalDMZ&df%5Bid%5D=DSD_EAG_LSO_EA%40DF_LSO_NEAC_UNEMP_MIGR&df%5Bag%5D=OECD.EDU.IMEP&dq=AUS%2BAUT%2BBEL%2BCAN%2BCHL%2BCOL%2BCRI%2BCZE%2BEST%2BFRA%2BDEU%2BGRC%2BHUN%2BIRL%2BISR%2BITA%2BLVA%2BLTU%2BLUX%2BMEX%2BNLD%2BNZL%2BNOR%2BPOL%2BPRT%2BSVK%2BSVN%2BESP%2BSWE%2BCHE%2BGBR%2BUSA%2BOECD%2BBGR%2BROU.M%2BF.Y25T34%2BY35T44%2BY55T64%2BY45T54%2BY25T64.ISCED11A_0T2%2BISCED11A_3_4%2BISCED11A_5T8...._Z%2BFB%2BNB._Z%2BY_LT16%2BY_GE16.....OBS...A3&lom=LASTNOBSERVATIONS&lo=1&pd=2023,2023&to%5BTIME_PERIOD%5D=true&lb=id&vw=ov.

R Core Team. 2021. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. https://www.R-project.org/.